

INFORMATICS PRACTICES (IP) - REPEATED BOARD TOPICS

Quick Revision Guide for High Scoring Sections

1. Computer Networks

Guaranteed 5 Marks

- **Server Placement:** Suggest the block with the **maximum number of computers**.
- **Cabling:** Use **Star Topology** (connect all blocks to the server building).
- **Repeaters:** Suggest if the distance between buildings is $> 70\text{-}100$ meters.
- **Hub/Switch:** Needed to connect computers *within* a single block.
- **Network Type:** **LAN** (within building), **MAN** (within city), **WAN** (between cities).

2. Data Visualization (Matplotlib)

4 Marks (Fill-in-blanks)

- `import matplotlib.pyplot as plt` (Library import)
- `plt.plot(x, y)` for Line Charts | `plt.bar(x, y)` for Bar Charts.
- `plt.xlabel()`, `plt.ylabel()`, and `plt.title()` for labeling.
- `plt.show()` and `plt.legend()` to display the graph.

3. SQL Functions & Queries

8-10 Marks Total

- **Math:** `MOD(n, m)` (remainder) and `ROUND(n, d)` (decimal rounding).
- **String:** `LENGTH()`, `SUBSTR(str, start, len)`, and `INSTR(str, sub)`.
- **Aggregates:** `SUM`, `AVG`, `MIN`, `MAX`, `COUNT`. Always check if `GROUP BY` is needed.
- **DDL:** `CREATE TABLE table_name (col1 type PRIMARY KEY, ...)`.

4. Python Pandas (DF & Series)

Core Coding

- **Creation:** Know how to convert a **Dictionary** or **List** into a DataFrame.
- **Attributes:** `df.columns` , `df.shape` , `df.size` .
- **Methods:** `df.rename()` , `df.drop()` , `df.head()` , and `df.tail()` .
- **Slicing:** Accessing rows/columns using `df.loc[]` or boolean conditions.

5. Societal Impacts & Ethics

Theory Section

- **E-waste:** Environmental hazards of toxic materials and the 3Rs (Reduce, Reuse, Recycle).
- **IPR:** Copyright (creative work), Trademark (logos), and Plagiarism (theft of ideas).
- **Cybercrime:** **Phishing** (fake emails) and **Cyberbullying** (harassment).
- **Footprints:** **Active** (updates/posts) vs **Passive** (IP address/browsing history).

IT (802) Add-on: Prepare Java Loops, Constructors, Try/Catch blocks, and Employability Skills (Communication/Self-Motivation).

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